

COURSE SYLLABUS

CSC13009 – Mobile Application Development

1. GENERAL INFORMATION

Course name:	Mobile application development
Course name (in Vietnamese):	Phát triển phần mềm cho thiết bị di động
Course ID:	CSC13009
Knowledge block:	Specialty
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	0
Prior course:	No
Instructors:	Toan-Thinh Truong

2. COURSE DESCRIPTION

This subject provides all students with the overview of developing mobile application development, some knowledges relating to main components in this area (some background, environment development & tools, user interface designs, libraries, test and deploy the application on smart phones). Some topics in this subject include: User interface design; application life cycle management; storage & data access; multi-thread processing; data-sharing among applications; service management; inter-process communication; using map service & GPS; some functions like call & message, animation, multimedia, and cryptography. Operating system in smart phones presented in this subject may be changed with the technology trends of the world and Vietnam. Now, this subject uses Android/iOS to demonstrate how to develop the application in mobile devices.

3. COURSE GOALS

At the end of the course, students can

ID	Description	Program LOs
G1	Work in group and collaborate to develop the mobile applications	2.2, 2.3.1
G2	Know and explain the English terminologies in this area	2.4.3, 2.4.5
G3	Explain basic conceptions, responsibility, task and ethical issues... in this area	1.4, 3.3
G4	Classify the background and environment of mobile application development	1.3.6, 1.4
G5	Understand and use some basic techniques in development of mobile application	5.1.1, 5.1.3, 5.2.1, 5.2.2, 5.3.1, 6.1.1
G6	Build a small-scale application personally	5.1.3, 5.2.2, 5.3.2, 6.1.2
G7	Use external tools to support the development in Android	1.3.6

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Build, organize, operate and manage the group	I, T
G1.2	Discuss with all topics in this subject	U
G1.3	Analyze, summarize and write technical reports as a group or in person	I, T
G2.1	Know and understand English terminologies of the subject	I

G2.2	Reading materials relating the lessons	I
G3.1	Explain the basic conceptions in this subject: market, backgrounds, environments, tools, libraries, ...	I, T
G3.2	Know the role, responsibility and ethical issues when working in this area	I
G3.3	Know how to update the knowleges, self-learn, develop and adapt new technologies	I
G3.4	Know how to start up in this area	I
G4.1	Differentiate between backgrounds and environments in this subject	I, T
G5.1	Efficiently design the mobile application in smart phones	I, T
G5.2	Understand the management of the life cycle in mobile application	I, T, U
G5.3	Build the multi-thread application	I
G5.4	Know how to interact with the smartphone's sensors	I, T
G5.5	Know how to store and access the data in smartphone	I, T, U
G5.6	Know how to integrate multimedia objects in applications	I, T, U
G5.7	Know how to develop location-based applications & web service	I, T, U
G6.1	Build a simple application systematically	I, T, U

G7.1	Use some simple tools and apply support technologies (for example: SQLite administrator, Drawdroid...)	I, T, U
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5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduce & overview the mobile device's market	G2.1, G2.2, G3.1, G3.2, G3.3, G3.4, G4.1	Lecturing Discussion Demonstration	H#1 Videos about Android/iOS
	Some important features in background and development environment	G1.2, G2.1, G2.2, G3.1, G3.3, G3.4		
	Analyze "Hello World" application	G1.2, G1.3, G2.1, G2.2, G3.1, G4.1		
2	Program the interface and manage the event	G1.2, G2.1, G2.2, G5.3, G6.1	Lecturing Demonstration Discussion Self-reading	H#2
	Some interface layouts	G1.2, G2.1, G2.2, G5.3, G6.1		
	Some single interface layouts	G1.2, G2.1, G2.2, G5.3, G6.1		
	Some complex interface layouts	G1.2, G1.3, G2.1, G2.2, G5.3, G6.1		

	Menu	G1.2, G2.1, G2.2, G5.3, G6.1		
3	Mange the application's life cycle	G1.2, G2.1, G2.2, G5.2, G6.1	Lecturing Discussion Demonstration	H#3
4	Mange the application with many screen, and pass the parameters between them	G1.2, G1.3, G2.1, G2.2, G5.3, G6.1	Lecturing Discussion Demonstration	H#4
5	Store and accesss data	G1.2, G1.3, G2.1, G2.2, G5.5, G6.1	Lecturing Discussion Demonstration	H#5
6	Process the multi-thread applications	G5.3, G6.1	Midterm Exam Lecturing Discussion Demonstration	H#6
7	Communicate among the applications Storage with file, xml, json, rss SQLite	G1.2, G1.3, G2.1, G2.2, G5.3, G6.1	Lecturing Discussion Demonstration	H#7
8	Multimedia (video & audio) Notification & Alarm	G1.2, G1.3, G2.1, G2.2, G5.3, G6.1	Self-reading Demonstration	H#8
9	Exploit the web services and API	G1.2, G1.3, G2.1, G2.2, G5.3, G5.7, G6.1	Lecturing Discussion Demonstration	H#9

10	Exploit the map service, location & GPS	G1.2, G1.3, G2.1, G2.2, G5.3, G5.7, G6.1	Lecturing Self-reading Demonstration	H#10
11	Seminar			

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
	Homework: HW1	- Survey and present how to + install Android studio and make a simple project “Hello World” + install Xcode on macOS and make a simple project “Hello World” - Submit a writing reports in English/Vietnamese (PDF)		2%
	Homework: HW2 → HW10	HW2 → HW10: practicing based on knowledge taught in class		18%
A2	Midterm Projects			30%

A21	Project	Midterm project		30%
A23	Final Projects			50%
A31	Seminar	Seminar the project schedule in theory class		10%
A32	Final project	Test with lab assistants		40%

7. RESOURCES

Textbooks for Android

- Unlocking Android - A Developer's Guide, W. Frank Abelson, Charlie Collins, and Robi Sen. Manning Pub., 2009
- Android Application Development: Programming with the Google SDK, Rick Rogers, John Lombardo, Zigurd Mednieks, Blake Meike. O'Reilly Pub., 2009.

Textbooks for iOS

- Molly K. Maskrey, **Beginning iPhone Development with Swift 5, 5th Edition, 2019**
- Apple Inc., **The Swift Programming Language**, 5rd Edition, 2019
- Gary Elmer, **Programming iOS 14 Using Swift UI_Get Started With Swift 5 and Xcode**, 2020.

Others

- Android Developer's Guides – available at: <http://developer.android.com/>
- Apple Developer: <http://developer.apple.com/>
- Tool: Android studio IDE, Xcode

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.